



7 μ J/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks

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ABSTRACT

Dynamic vision sensor (DVS) cameras enable energy-activity proportional visual sensing by only propagating events produced by changes in the observed scene. Furthermore, by generating these events asynchronously, they offer μ s-scale latency while eliminating the redundant data transmission inherent to classical, frame-based cameras. However, the potential of DVS to improve the energy efficiency of IoT sensor nodes can only be fully realized with efficient and flexible systems that tightly integrate sensing, processing, and actuation capabilities. In this paper, we propose a complete end-to-end pipeline for DVS event data classification implemented on the Kraken parallel ultra-low power (PULP) system-on-chip and apply it to gesture recognition. A dedicated on-chip peripheral interface for DVS cameras aggregates the received events into ternary *event frames*. We process these video frames with a fully ternarized two-stage temporal convolutional network (TCN). The neural network can be executed either on Kraken's PULP cluster of general-purpose RISC-V cores or on CUTIE, the on-chip ternary neural network accelerator. We perform extensive ablations on network structure, training, and data generation parameters. We achieve a validation accuracy of 97.7% on the DVS128 11-class gesture dataset, a new record for embedded implementations. With in-silicon power and energy measurements, we demonstrate a classification energy of 7 μ J at a latency of 0.9 ms when running the TCN on CUTIE, a reduction of inference energy by 67 $\times$ when compared to the state of the art in embedded gesture recognition. The processing system consumes as little as 4.7 mW in continuous inference, enabling always-on gesture recognition and closing the gap between the efficiency potential of DVS cameras and application scenarios.

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1. Introduction

The exponential proliferation of small-scale sensing systems such as Internet of Things (IoT) sensor nodes and wearable devices has coincided with a revolution in data processing, where machine learning (ML) algorithms such as CNNs and transformers have dramatically improved the state of the art in fields such as computer vision, natural language processing or biomedical signal processing [1–4]. As many applications involving the collection of data demand for their interpretation and reaction to the result, these developments are naturally symbiotic. However, the approach of transmitting raw data to cloud servers to be processed with powerful and compute-intensive ML model does not scale to the enormous amounts of data collected and conflicts with the constraints of battery-powered, sensor-driven

applications requiring real-time performance: wireless communication is power-intensive and incurs communication latency, and the transmission of raw sensor data raises privacy and security concerns.

This conflict is addressed by the emerging research field of edge AI, which aims to enable efficient and accurate ML-based processing of sensor data on edge sensor nodes. These systems, typically built around Microcontrollers (MCUs), operate under stringent power, memory, and compute resource constraints. A key algorithmic technique in edge AI is *quantization* of neural networks, where parameters and intermediate activations are represented in low-bitwidth data formats, reducing a model's memory and storage footprints. With hardware support for low-precision arithmetic through specialized extensions to general-purpose instruction set architectures (ISAs) [5] or dedicated accelerators, quantization also increases computational throughput and efficiency, leading to lower energy consumption per inference.

However, to optimize a system's power consumption and performance, the entire pipeline from the sensor to the reaction to

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